

DOE Title Page

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

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Request for Application (RFA) Number: DE-FOA-0003612

Phase I

Primary Challenge and Focus Area: 21-A Artificial Intelligence in Fluid Flow for Energy
Components and Technologies | Physics-Informed AI for Complex Flow Modeling

Senior/Key Personnel for Prime Applicant/Lead Institution and All Partner Institutions:

Florida State University (Lead): Mark Sussman, Kourosh Shoele

Sandia National Laboratories: Marco Arienti

Summary Budget Information for prime applicant/lead institution and all partner institutions:

Budgets are Year 1 budgets

Florida State University (Lead) : 296096

Sandia National Laboratories: 150000

Total: 446096

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

1. Background/Introduction

Relevance to DOE Genesis Mission: This proposal responds directly to “The Genesis Mission: Transforming Science and Energy with AI,” specifically Topic 21, “Artificial Intelligence in Fluid Flow for Energy Components and Technologies,” and Focus Area 21A, “Physics-Informed AI for complex fluid phenomena.”

We seek support to develop AI-enhanced numerical methods for atomization and spray processes that are central to combustion in aircraft engines [1] and rotating detonation engines used in rocket propulsion [2, 3]. These flows are especially challenging because they involve strongly coupled, multiscale interfacial dynamics that are difficult to resolve efficiently without losing predictive fidelity.

Our approach places AI within an efficient computational framework that uses high-fidelity simulations and targeted experimental data to train sub-scale models that generalize across atomization and spray problems. In direct alignment with Topic 21A, we will: **(1)** assimilate data from high-fidelity simulations and/or experiments into our existing multiphase-flow methodology [1–9]; and **(2)** use AI to learn dynamic, non-local corrections that recover the gap between under-resolved and high-fidelity simulations and/or experiments.

The overarching goal is to reduce the need for repeated high-fidelity simulations while maintaining physical consistency and predictive capability. More broadly, this work can strengthen predictive modeling for fuel injection, combustion, boiling, evaporation, condensation, and advanced propulsion systems, while advancing adaptive digital environments that tightly couple models, data, and computation.

Proposed Research in the context of the current state-of-the-art: Multiphase flows with *sharp, dynamically evolving interfaces* play a central role in energy conversion and transport systems, including fuel injection, spray combustion, rotating detonation engines, and phase-change-driven thermal processes [1–5, 10]. These flows are governed by *tightly coupled nonlinear interactions* among turbulence, interfacial forces, thermodynamics, and material response across a wide range of spatial and temporal scales, and are critical for identifying distinct mass and momentum processes at evolving interfaces. Their accurate prediction remains **one of the most challenging problems in computational science**.

Over the past several decades, semi-implicit **sharp-interface methods** have matured into a rigorous and reliable framework for modeling such systems. Foundational work by [11] established level set methods for tracking evolving interfaces in multiphase flow. Later advances produced conservative formulations [12] that significantly improved *mass conservation and robustness*, especially in the presence of topology change.

A major advance in this area is the **moment-of-fluid (MOF)** [13–15] and cell-integrated semi-Lagrangian **compressible MOF framework developed by Jemison, Sussman, and Arienti** [16]. The compressible multiphase formulation developed by Jemison et al. introduced a unified method that conserves mass, momentum, and energy while capturing shocks and material discontinuities without Riemann solvers. The material interfaces in [16] are represented sharply

while preserving volume to machine precision in zero Mach number regions. Our approach enables simulations across compressible and incompressible fluids through asymptotic-preserving semi-implicit pressure updates. Further developments demonstrated accurate treatment of *multi-material junctions* and surface tension effects in complex geometries [2, 3, 5–8, 17]. However, the cost of resolving these multiscale interfacial processes at the fidelity required for predictive simulation remains high, making AI-enhanced modeling a natural and increasingly necessary next step.

This challenge is especially evident in high-fidelity DNS and LES studies of fuel injection and atomization, which reveal the *extreme computational cost* of resolving interface breakup, turbulence interaction, compressibility effects, and sharp mass transfer at fluid interfaces under realistic engine conditions. Such cost limits their usefulness for engineering design, where rapid turnaround over broad parameter spaces is essential. For liquid-jet-in-crossflow (JICF) atomization, even adaptive mesh refinement provides only limited practical benefit [10], because of the strong coupling between chaotic, bulk, gas-phase motion and liquid breakup. This motivates AI-enabled modeling and solution strategies that preserve predictive fidelity while overcoming fundamental computational bottlenecks.

One possible avenue of improvement we considered, as well as others [7, 18], was to replace costly geometric operations in multiphase flow algorithms with surrogate AI-generated models. The key principle is that surrogate models must not compromise on the fidelity and stability of the numerical algorithm. In particular, for volume tracking, or moment of fluid methods, it is essential that the “linearity preserving” and “volume preserving” properties of a multiphase flow system not be violated [13, 14, 19–21]. Yet, what was discovered by us [7], and also mentioned in [18] is that these AI generated surrogate algorithms, at best, provide a marginal benefit. We also bring attention to the following articles which give evidence that surrogate AI models will never beat finely tuned optimized algorithms [22–25].

Another possible direction is to pursue meshless methods such as PINN [26, 27] or SPH [28, 29]. However, these approaches, and pure meshless methods more generally, do not inherently preserve volume. This is a fundamental limitation, because in the low-Mach-number limit the fluid volume should remain conserved. Notably, the volume error reported for PINN in [26] was *worse* than that of classical level set methods. In this sense, they revisit the main criticism of early level set methods: lack of strict volume conservation.

It is proposed herein an alternative solution by pursuing two tightly connected advances: (I) we shall augment our existing compressible multiphase flow algorithm [2, 3, 16] with a volume- and mass-preserving compressible multiphase-flow nudging capability for the assimilation of data from high-resolution, high-fidelity simulations and/or experiments [30]. (II) We shall then use the nudging correction generated in (I) to train a conditioned AI model to construct sub-scale models for under-resolved multiphase flows. The resulting sub-scale model is expected to generalize across a broad class of multiphase flow problems.

2. Project Objectives

The project objectives are twofold:

OBJ. 1: We will advance our existing compressible multiphase flow algorithm [2, 3, 16] by developing an AI-centric compressible multiphase-flow nudging capability. The proposed nudging algorithm will enable volume- and mass-preserving “interface nudging” tailored to the complex atomization

and spray problems considered here. Specifically, the nudging term will take the form $-\frac{p_s}{\rho}\nabla H$, where p_s is a nudging stress acting normal to the interface and is proportional to the difference between the simulated level set function and an assimilated level set function [30].

OBJ. 2: We will then use the nudging correction generated in (obj. 1) to train an AI model with regularization terms to derive sub-scale models for underresolved multiphase flows. The resulting model is expected to generalize across a broad range of multiphase flow regimes.

Our targeted canonical problems are: (a) atomization of a fuel jet in crossflow (JICF), which is directly relevant to combustion in aircraft engines [1]; and (b) the shock cylinder/drop problem, which is relevant to rotating detonation engines used in rocket propulsion [2, 3].

A successful Phase I outcome will create a clear pathway to **Phase II** by delivering AI foundations needed for extension to more realistic multiphase flow environments. This next step would target full three-dimensional, engine-relevant problems for predictive design and analysis.

3. Proposed Research and Methods; Physics-Constrained Data Driven Model

The proposed research builds on prior work by PIs **Sussman, Shoele, and Arienti** [5–8, 16], particularly the *cell-integrated semi-Lagrangian framework* and *moment-of-fluid (MOF) interface representation*, which enable accurate tracking of deforming interfaces with efficient treatment of compressibility and stiffness. The governing equations enforce conservation of mass, momentum, energy, and species, with extensions to phase change and related interface physics, including stress balance, mass-transfer-induced velocity jumps, and thermodynamic equilibrium.

Within this foundation, we propose a *physics-constrained, solver-embedded learning framework* in which high-fidelity computational and/or experimental data are assimilated, via nudging, into a coarse-grid two-phase-flow simulation, while neural networks learn unresolved interfacial and near-interface corrections from the resulting nudging source terms. Unlike PINN-style approaches that approximate the full solution field, this framework retains the conservative MOF-based solver, preserving sharp interface representation, strict mass conservation, and compatibility with established multiphase algorithms. AI therefore plays a focused role: learning localized corrections embedded directly into the governing equations rather than replacing the underlying physics-based solver.

The Phase I effort proceeds in three connected steps.

Task 1: we will implement a compressible multiphase nudging capability that assimilates interface, velocity, and temperature information from high-fidelity simulations and/or experiments into a coarse-grid solver while preserving volume and mass.

Task 2: we will develop the data-handling and restriction machinery needed to map fine-grid information onto the coarse-grid representation, thereby producing dynamically aligned coarse/high-fidelity state pairs.

Task 3: we will use the nudging correction generated by this alignment procedure to train a neural network that provides sub-scale corrections for underresolved multiphase flows. In this way, the AI model is trained not on arbitrary field mismatch, but on physics-based corrections extracted from an interface-resolved solver.

A central issue is that independently evolved coarse-grid and high-fidelity simulations diverge rapidly because interfacial errors amplify nonlinearly. As a result, the raw difference between the two

solutions does not provide a clean learning target for unresolved physics. Instead, the coarse solution is periodically assimilated toward the restricted high-fidelity state, $Q_{hf \rightarrow c}^{n+1}$, and the aligned coarse state is then advanced by one coarse-grid time step. The discrepancy between this uncorrected update and the corresponding high-fidelity evolution defines the target correction that the neural network is trained to learn. This strategy ensures that the learned model captures missing interfacial and subgrid physics, rather than compensating for accumulated trajectory drift. Let Q_a^n denote the assimilated coarse state at time t_n , and let $Q_{c,unc}^{n+1} = G_{\text{MOF}}(Q_a^n)$ denote the corresponding uncorrected coarse update. The target correction is

$$F_{\text{corr}}^n = \frac{Q_{hf \rightarrow c}^{n+1} - Q_{c,unc}^{n+1}}{\Delta t}.$$

To implement the learned closure, we will construct training data directly from the nudging-based alignment procedure so that the neural network learns the one-step correction needed to recover high-fidelity interfacial dynamics from an underresolved coarse-grid update. The network inputs will include coarse-grid flow variables, thermodynamic fields, and reconstructed interface features—including volume fraction, centroids, interface normals, curvature, and a narrow-band interface mask—while the outputs will be localized correction terms added to the governing equations for momentum, energy, species, and, where needed, interface-normal dynamics. A neural network $\mathcal{N}_\theta$ will be trained to map coarse-grid flow and interface features, $z^n = \Phi(Q_a^n, \Gamma_a^n)$, to F_{corr}^n using a loss of the form $\mathcal{L}(\theta) = \sum_n \|W \cdot (\mathcal{N}_\theta(z^n) - F_{\text{corr}}^n)\|^2 + \mathcal{R}(\theta)$, where the weighting W and regularization $\mathcal{R}(\theta)$ are concentrated near the interface. As a practical Phase I step, we will use a localized residual/U-Net model for narrow-band correction learning and evaluate a neural-operator extension, *e.g.* U-FNO, to capture longer-range coupling and improve generalization across operating conditions.

Because the dominant coarse-grid errors arise in mixed cells and in a narrow band surrounding the reconstructed interface, both supervision and regularization are concentrated in these regions. The loss is weighted to emphasize interfacial discrepancies, while the regularization is designed to keep the learned correction localized, smooth, bounded, and geometrically consistent with the interface. In practice, this includes penalties that suppress spurious bulk corrections, reduce nonphysical oscillations, limit overly large predicted terms, and, for interfacial force corrections, promote alignment with the reconstructed interface normal. These constraints improve stability and keep the learned model focused on the unresolved interfacial physics most critical to predictive simulation.

When high-fidelity data are available, the nudging procedure aligns the coarse solution with reference dynamics at a fraction of the computational cost; when material interfaces are nudged, volume and mass preservation are enforced using the algorithm of Dommermuth et al. [30]. The resulting framework provides a clear and concise Phase I pathway: develop the assimilation capability, construct the learning target and neural-network correction model, and then verify and validate the integrated method on canonical atomization and spray problems. In this way, AI is used to extend the predictive reach of high-fidelity multiphase simulation while preserving the conservative structure and interface fidelity of the underlying solver.

4. Milestones and Deliverables

- 1. (Sussman, Months 1–3):** Enhance the current compressible MOF algorithm [2, 3, 5, 16] by adding a nudging-based data-assimilation capability. Volume and mass conservation during nudging will be ensured using the algorithm of Dommermuth et al. [30]
- 2. (Arienti, Months 1–3):** Develop routines to import high-resolution data [2, 3] into actively running coarse-grid simulations and construct the restriction operation $Q_{hf \rightarrow c}^{n+1}$.
- 3. (Shoele, Months 1–3):** Develop the neural-network model that minimizes the discrepancy between predicted and target corrections, $\mathcal{L}(\theta) = \sum_n \|W \cdot (\mathcal{N}_\theta(z^n) - F_{\text{corr}}^n)\|^2 + \mathcal{R}(\theta)$, with weighting W and regularization $\mathcal{R}(\theta)$ concentrated near material interfaces.
- 4. (Arienti, Sussman, Shoele, Months 4–6):** Integrate Tasks 1–3, including nudging-based assimilation, AMReX data management, and AI-generated sub-scale models. Perform diagnostic testing, verification, and validation on 2D shock-cylinder atomization and spray problems.
- 5. (Arienti, Sussman, Shoele, Months 7–9):** Extend Task 4 to fully 3D problems, with corresponding testing, verification, and validation.

5. Decision Gate Metrics

We will train the neural-network subscale models on the 2D/3D water and Galinstan shock-cylinder interaction problems shown in Figures 4, 6, and 7 of [3]. The effective fine-grid resolution for the high-fidelity simulation is $3072 \times 2304 \times 768$ (3D), while the corresponding low-fidelity resolution is $384 \times 288 \times 96$. After training, we will rerun the water and Galinstan cases using the learned model in place of direct high-fidelity assimilation and compare the resulting interfaces among: (a) coarse-grid assimilated results, (b) coarse-grid model-corrected results, and (c) high-fidelity results. We will then introduce two additional cases with varied Weber and Reynolds numbers to assess generalization. Interface errors will be quantified using the symmetric difference error [15].

6. Data Sources and Models

The current AMReX-based compressible multiphase solver used to generate Figure 3 of [1], Figures 4 and 6 of [3], and Figures 5 and 11a of [2] is maintained on GitHub at https://github.com/msussman42/amrex_implicit_interfaces.git. The data for Figure 3 of [1] and Figure 4 of [3] can be regenerated within 30 days. The remaining high-fidelity datasets are available through Marco Arienti on national laboratory computing resources.

7. Anticipated Phase I Outcomes and National Impact

Phase I will deliver an integrated AI-enhanced workflow for improved prediction of atomization and spray dynamics using limited high-fidelity simulation and/or experimental data. The key outcome will be a physics-constrained framework that combines compressible multiphase nudging, solver alignment, and learned sub-scale corrections within a conservative sharp-interface solver. This will lift the reliance on repeated high-fidelity calculations while preserving physical consistency and predictive fidelity. The broader impact is significant: improved predictive capability for atomization, spray breakup, and evaporation can strengthen computational design and analysis for energy-relevant multiphase systems. A successful Phase I will provide the foundation for Phase II, where the methodology will be extended to more realistic three-dimensional and operating conditions.

Appendix 1: Bibliography and References Cited

phase I

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

PI: Mark Sussman, Department of Mathematics, Florida State University

CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering

CO-PI: Marco Arienti, Sandia National Laboratories

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Appendix 2: Facilities and Other Resources

phase I

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

PI: Mark Sussman, Department of Mathematics, Florida State University

CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering

CO-PI: Marco Arienti, Sandia National Laboratories

Laboratory Resources: The FAMU-FSU College of Engineering performance site facilities used in this project are located within the innovation park at the joint college of engineering in Tallahassee. CO-PI Shoele is a member of Florida Center for Advanced Aero-Propulsion (FCAAP) and has a 270 square foot lab equipped with computers. Office space for graduate students are provided by FCAAP.

Computer (Shoele): CO-PI Shoele’s laboratory is equipped with over a dozen PC workstations. Key software includes Tecplot, GridGen, Intel compilers. The department of Mechanical Engineering and FCAAP provide the software as well as printing facilities.

Computer (Sussman): PI Sussman has access in the math department to 8 4-core computers, 1 32 core machine, and 1 8-core machine. The 8-core machine has an Nvidia GPU card in it.

Computing Facilities (FSU-FAMU): There are a number of major computing facilities at FSU-FAMU that are available to the PIs Sussman, Shoele, and students in FSU-FAMU for this project. 1) HPC Cluster: 647 node, 10772 core, cluster with CUDA GPU capability attached to 256 TB Panasas file system that is mounted on all of the HPC compute nodes over a dedicated 1Gbps network. 2) Spear cluster: 88 x86 64-bit process cores linked together by a QDR Infiniband network. Eight of the nodes also connect to a PCI chassis equipped with 16 NVIDIA Tesla gpGPU cards (m2050) for a total 7,168 GPU cores for a peak system performance of 19.8 TFLOPS. 3) CAPS Real Time Digital Simulator (RTDS): A parallel-processing machine (performance at over 110 GigaFLOPS) with “14- rack” system consisting of 380 parallel processors and employing optimized EMTP-based nodal network simulation methods.

Computing Facilities (Sandia National Laboratories): Sandia National Laboratories’ HPC Production Clusters for Institutional Programs provide computing resources through several high-bandwidth platforms (CPU- and GPU-based) optimized for scientific computing. In addition, Sandia offers state-of-the art file managing options for fast metadata operations and large data storage.

Appendix 3: Equipment

phase I

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

PI: Mark Sussman, Department of Mathematics, Florida State University

CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering

CO-PI: Marco Arienti, Sandia National Laboratories

The primary equipment that will be used for this project will be:

- 1.** The FSU HPC Cluster will be used for generating high fidelity test data to be compared with the “low fidelity + AI generated model.” Also, the FSU HPC Cluster will be used in the “AI model generation stage.”
- 2.** The math department 32 core and 8 core machines will be used for algorithmic development as well as the workstations in Shoele’s lab.
- 3.** At any given time, one of the Sandia National Laboratories’ HPC Production Clusters will be available for the purpose of importing existing high fidelity data into running coarse-grid simulations.

**Appendix 6: Transparency of Foreign Connections Disclosure and Certification
phase I**

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

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This format is provided for the convenience of the entity providing the disclosure and certification; however, the entity is not required to use this specific format. If another format is used, the signatory must include the same substantive information, a signature, date, and the certification statement provided in Section 3 below.

DEPARTMENT OF ENERGY TRANSPARENCY OF FOREIGN CONNECTIONS DISCLOSURE AND CERTIFICATION

For the Authorized Official of the entity (required for both the recipient and subrecipients)

The following information is required as part of the Department of Energy's (DOE) research, technology and economic security due diligence review process. Each entity must disclose the following information as it relates to its respective entity. Include a separate disclosure and signed certification for each entity at the recipient and subrecipient level.

All items below require a response, unless exempt by entity type. If supplemental documentation is provided, identify the applicability to each item below. Supplemental information, including links to financial statements or annual reports, is not a replacement for responses. If you have nothing to report for a specific item, include "Nothing to Report" for that item.

Responses to disclosure questions may contain trade secrets or commercial or financial information that is privileged or confidential and is exempt from public disclosure. Such information shall be used or disclosed only for evaluation purposes or in accordance with an award between the submitter and the United States Government. See Appendix B for information on how to identify confidential information.

Save each TFC in a single file using the following convention for the title:
"BusinessSensitive_AwardNumber_EntityName(Prime or Sub)_TFC".

DOE reserves the right to request additional or clarifying information based on the information submitted.

Disclosure exceptions by entity type:

- DOE National Laboratories and domestic government entities are not required to respond to the Transparency of Foreign Connections disclosure.
- Institutions of higher education are only required to respond to items with an asterisk symbol (*).
- The applicability of the disclosure requirements is determined by the entity type. Regardless of whether the recipient is exempt, the subrecipient(s) must provide these disclosures unless the subrecipient is also exempt.

Questions: Contact rtesinfo@hq.doe.gov

*Section 1. Identifying Information

a. *Complete Legal Name of the Entity	Florida State University
b. *Entity's Unique Entity ID (UEI)	JF2BLNN4PJC3
c. *Entity's Website Address (link)	https://www.research.fsu.edu/research-offices/sra/
d. *Entity's Mailing Address	Sponsored Research Administration, 874 Traditions Way, Third Floor, Tallahassee, FL 32306-4166
e. *Entity's Physical Address	Sponsored Research Administration, 874 Traditions Way, Third Floor, Tallahassee, FL 32306-4166
f. *Entity's Business Point of Contact (POC) Name	Alexander Karabedian
g. *POC Phone Number	850-644-5260
h. *Project Title	Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions
i. *Proposal # or Award #	DE-FOA-0003612

Section 2. Disclosure of Foreign Connections

1. *Malign Foreign Talent Recruitment Program

*Is any principal investigator, project manager, or other covered individual, or business owner, officer party to a Malign Foreign Talent Recruitment Program (MFTRP)? ☐ Yes ☒ No

*If yes, provide the following information: Full name, position, role in the project, name of the MFTRP, and dates of participation.

2. Ownership Structure

- a. Provide a complete capitalization table for your entity, including all equity interests (including LLC and partnership interests, as well as derivative securities). Include both the number of shares issued to each equity holder, as well as the percentage of that series and of all equity on a fully diluted basis. Identify the principal place of incorporation (or organization) for each equity holder. If the equity holder is a natural person, identify the citizenship(s).

- If the recipient or subrecipient is a publicly traded company, provide the above information for shareholders with an interest greater than five percent.
- b. For any owner/investor identified that is an investment fund utilizing a general partner structure, identify: 1) the general partner or equivalent individual/entity; and 2) the place of incorporation and principal place of business if the general partner is an entity OR citizenship(s) if the general partner is a natural person.
- c. Provide an entity chart (including ownership percentages and officers) to illustrate the relationship between your entity, your entity's immediate parents, intermediate parents, ultimate parents, and any subsidiaries.
- i. For your entity's ultimate parent, please identify all owners with greater than a five percent ownership interest.
 - ii. For entities, please identify the locations of incorporation and principal places of business. For individuals, please identify countries of citizenship.
- d. Are any entities, owners, or investors within your entity's ownership chain identified on the Bureau of Industry and Security Entity List (search via [Consolidated Screening List](#) and select "Entity List" for "Source"), [Department of Defense 1260H List](#), and/or Department of the Treasury [Office of Foreign Assets Control Treasury's Specially Designated Nationals list](#)? ☐ Yes ☐ No

If yes, identify entity, owner or investor and the specific list.

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- e. Provide a summary table identifying all rounds of financing, the purchase dates, the investors for each round, and all of the associated governance and information rights obtained by investors during each round of financing.
3. Is your entity wholly or partially owned by an entity incorporated or otherwise formed in a foreign country of concern or foreign state-owned entity?
- ☐ Yes ☐ No

If yes, disclose the entity name, full address, and percentage of ownership.

4. Is there a parent company, joint venture, or subsidiary, of your entity that is based in or receives funding from any foreign country of concern? ☐ Yes ☐ No

If yes, disclose the entity name, full address, applicant or awardee relationships (i.e., parent company, joint venture, subsidiary, etc.) of each entity based in, or funded by, any foreign country of concern.

5. Does your entity have any current or pending contractual or financial obligation or other agreement specific to a business arrangement, or joint venture-like arrangement with an entity based in, or funded by, any foreign country of concern? ☐ Yes ☐ No

If yes, proceed to question 5a. If no, proceed to question 6.

5a. Is the obligation, agreement, or arrangement connected to the project identified in Section 1?

☐ Yes ☐ No

If yes, disclose the name of each enterprise or foreign entity, type of obligation, agreement, or arrangement (i.e., contractual, financial, or other), description of obligation, agreement, or arrangement, and the foreign state(s) and/or the country of the foreign entity (or entities).

6. Does your entity have any venture capital or institutional investment? ☐ Yes ☐ No

If yes, proceed to question 6a. If no, proceed to question 7.

6a. Does the investing entity have a general partner or any other individual holding a leadership role who has a foreign affiliation with a foreign country of concern?

☐ Yes ☐ No ☐ Unable to determine

If yes or unable to determine, disclose the venture capital or institutional investing entity's name, the percentage of ownership obtained by the investing entity, and the type of investment (i.e., equity, debt, or combination of equity and debt).

7. *During the previous 5-year period, did your entity have any technology licensing or intellectual property sales or transfers to a foreign country of concern? ☒ Yes ☐ No

If yes, proceed to question 7b. If no, proceed to question 8.

*7b. Was the underlying technology in the same technology area (e.g., batteries, biotechnology, grid, energy generation and storage, advanced computing) as the project identified in Section 1? ☐ Yes ☒ No

*If yes, disclose the name, address, and country, of the institution or entity that licensed, purchased, or received the technology or intellectual property.

8. Is there any foreign business entity, offshore entity, or entity outside the United States related to your entity? ☐ Yes ☐ No

If yes, disclose the entity name, relationship type (i.e., foreign business entity, offshore entity, entity outside the United States), description of the relationship to the applicant or awardee, and entity address and country.

9. Provide a complete list of all directors (and board observers), including their full name, shareholder affiliation, date of appointment, voting rights, citizenship, duration of term, as well as a description of observer rights as applicable. *This list may be submitted as an attachment.*

10. Does your entity, whether as borrower, surety, guarantor, or other, have any indebtedness, liabilities, or obligations to an entity based in, or funded by, any foreign country of concern? ☐ Yes ☐ No

If yes, disclose the name of the entity and the details of the obligation.

11. *Will the project identified in Section 1 include any of the following? ☐ Yes ☒ No

- Federal Acquisition Security Council (FASC)-prohibited unmanned aircraft systems or the operation thereof.
- Coded equipment where the source code is written in a foreign country of risk.
- Equipment from a foreign country of concern that will be connected to the internet or other remote communication system.
- Equipment used on the project after delivery where any entity or individual based in a foreign country of concern will have physical or remote access to any part of the equipment.

If yes, disclose the name and address of the supplier, and the item proposed for use on the project:

12. * Is the DOE funding for the project identified in Section 1 supporting the deployment of battery energy storage systems (BESS)? ☐ Yes ☒ No

If yes, please answer the following questions:

- a. Please identify the vendor/integrator of any battery energy storage systems (BESS) covered under the award. Provide the model name/number of the BESS if possible.

- b. Please identify the inverter manufacturer for the BESS covered under the award. If the inverter manufacturer is the same entity as the primary BESS vendor/integrator, please provide that confirmation.

- c. Please identify the supplier of the battery management systems (BMS) for the BESS covered under the award. If the BMS provider is the same entity as the primary BESS vendor/integrator, please provide that confirmation.

- d. Please identify the manufacturer of the battery packs included in the BESS covered under the award. If the battery pack provider is the same entity as the primary BESS vendor/integrator, please provide that confirmation.

- e. Please identify the manufacturer of the battery cells included in the BESS covered under the award. If the battery cell provider is the same entity as the primary BESS vendor/integrator, please provide that confirmation.



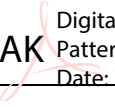
Section 3. Certifications

Individuals participating in a Malign Foreign Talent Recruitment Program are prohibited from participating in a DOE award. The recipient and the subrecipients must certify that 1) the covered individuals in their respective employment have been made aware of the Malign Foreign Talent Recruitment Program prohibition and 2) the individuals have complied with their responsibilities to annually certify they are not party to a Malign Foreign Talent Recruitment Program.

I am providing this disclosure and certification on behalf of the above-mentioned entity as part of my responsibilities as the entity's authorized official for the above-mentioned project. I certify to the best of my knowledge and belief that the information contained in this document is true, complete, and accurate.

I further certify that 1) the covered individuals in their respective employment have been made aware of the Malign Foreign Talent Recruitment Program prohibition and 2) the covered individuals have complied with their responsibilities to certify they are not party to a Malign Foreign Talent Recruitment Program.

I understand any false fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of material fact, may subject me to criminal, civil, or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. §§ 3729, 3733, and 3801-3812). I further understand and agree that (1) the statements and representations made in this document are material to DOE's funding decision; (2) in the event DOE issues an award, should I later learn during the life of the award that an individual participating in the project funded under the award is or is reasonably believed to be involved with a Malign Foreign Talent Recruitment Program, I must notify the cognizant DOE/NNSA Grants Officer within five business days on learning such information; and (3) I have an ongoing responsibility to update the disclosures provided above during the period of performance of the award if there is any change to a disclosure.

Authorized Official: Stacey Patterson	Title: Vice President for Research
Signature: Stacey Patterson/AK  Digitally signed by Stacey Patterson/AK Date: 2026.04.29 10:55:13 -04'00'	Date: 4/29/2026

Appendix A. Definitions

1. **Covered Individual.** The Department of Energy (DOE) has designated individuals who contribute in a substantive, meaningful way to the project proposed to be carried out with an award from DOE, at both the prime and subrecipient levels. Often, these individuals have doctoral or other professional degrees, although individuals at the master's or PhD level may be considered covered individuals if their involvement meets this definition. Consultants, graduate students, and those with a postdoctoral role also may be considered covered individuals if they meet this definition. Consult the Notice of Funding Opportunity or Award Terms and Conditions, as applicable, for the relevant definition of "covered individual."

2. **Foreign Country of Concern.** The term "foreign country of concern" means the People's Republic of China, the Democratic People's Republic of Korea, the Russian Federation, the Islamic Republic of Iran, or any other country determined to be a country of concern by the Secretary of State, and including countries of risk designated by the Department.

3. **Malign Foreign Talent Recruitment Program.** As defined in [P.L. 117-167, Section 10638 \(4\)](#), the term "malign foreign talent recruitment program" means:

(A) any program, position, or activity that includes compensation in the form of cash, in-kind compensation, including research funding, promised future compensation, complimentary foreign travel, things of non de minimis value, honorific titles, career advancement opportunities, or other types of remuneration or consideration directly provided by a foreign country at any level (national, provincial, or local) or their designee, or an entity based in, funded by, or affiliated with a foreign country, whether or not directly sponsored by the foreign country, to the targeted individual, whether directly or indirectly stated in the arrangement, contract, or other documentation at issue, in exchange for the individual –

- (i) engaging in the unauthorized transfer of intellectual property, materials, data products, or other nonpublic information owned by a United States entity or developed with a Federal research and development award to the government of a foreign country or an entity based in, funded by, or affiliated with a foreign country regardless of whether that government or entity provided support for the development of the intellectual property, materials, or data products;
- (ii) being required to recruit trainees or researchers to enroll in such program, position, or activity;
- (iii) establishing a laboratory or company, accepting a faculty position, or undertaking any other employment or appointment in a foreign country or with an entity based in, funded by, or affiliated with a foreign country if such activities are in violation of the standard terms and conditions of a Federal research and development award;
- (iv) being unable to terminate the foreign talent recruitment program contract or agreement except in extraordinary circumstances;
- (v) through funding or effort related to the foreign talent recruitment program, being limited in the capacity to carry out a research and development award or required to engage in work that would result in substantial overlap or duplication with a Federal research and development award;
- (vi) being required to apply for and successfully receive funding from the sponsoring foreign government's funding agencies with the sponsoring foreign organization as the recipient;

- (vii) being required to omit acknowledgment of the recipient institution with which the individual is affiliated, or the Federal research agency sponsoring the research and development award, contrary to the institutional policies or standard terms and conditions of the Federal research and development award;
- (viii) being required to not disclose to the Federal research agency or employing institution the participation of such individual in such program, position, or activity; or
- (ix) having a conflict of interest or conflict of commitment contrary to the standard terms and conditions of the Federal research and development award; and

(B) a program that is sponsored by—

- (i) a foreign country of concern or an entity based in a foreign country of concern, whether or not directly sponsored by the foreign country of concern.
- (ii) an academic institution on the list developed under [section 1286\(c\)\(8\) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 \(10 U.S.C. 2358 note; 1 Public Law 115–232\)](#); or
- (iii) a foreign talent recruitment program on the list developed under [section 1286\(c\)\(9\) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 \(10 U.S.C. 2358 note; 1 Public Law 115–232\)](#).

Appendix B. Identification of Confidential Information

If the response includes business sensitive information (e.g., commercial or financial information that is privileged or confidential), trade secrets, proprietary, or otherwise confidential information in their or otherwise confidential information, it is furnished to the federal government (government) in confidence and such information will be withheld from public disclosure to the extent permitted by law, including the Freedom of Information Act. In accordance with 10 CFR part 1004, DOE will make its own determination about the confidential status of the information and treat it according to its determination.

The response must be marked “confidential” and clearly and conspicuously identify the business sensitive, trade secrets, proprietary, or otherwise confidential information in accordance with the instructions below. Failure to comply with these marking requirements may result in the disclosure of the unmarked information under the Freedom of Information Act or otherwise. The government is not liable for the disclosure or use of unmarked information and may use or disclose such information for any purpose as authorized by law.

The first page of the response must be marked as follows and identify the specific pages business sensitive, trade secrets, proprietary, or otherwise confidential information:

Notice of Restriction on Disclosure and Use of Data:

Pages [list applicable pages] of this document may contain business sensitive, trade secrets, proprietary, or otherwise confidential information that is exempt from public disclosure. Such information shall be used or disclosed only for evaluation purposes or in accordance with a financial assistance agreement between the submitter and the government. The government may use or disclose any information that is not appropriately marked or otherwise restricted, regardless of source. [End of Notice]

In addition, (1) the header and footer of every page that contains business sensitive, trade secrets, proprietary, or otherwise confidential information must be marked as follows: “Contains Business Sensitive, Trade Secrets, Proprietary, or Otherwise Confidential Information Exempt from Public Disclosure,” and (2) every line or paragraph containing such information must be clearly marked with double brackets or highlighting. DOE will make its own determination about the confidential status of the information and treat it according to its determination.

Appendix 7: Letters of Commitment from Sandia National Laboratories and Florida State University

phase I

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

PI: Mark Sussman, Department of Mathematics, Florida State University

CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering

CO-PI: Marco Arienti, Sandia National Laboratories



SPONSORED RESEARCH ADMINISTRATION

Date: 4/29/26

Re: Genesis Proposal Submission in response to US Department of Energy Notice of Request for Application **DE-FOA-0003612**
Florida State University Principal Investigator: Dr. Mark Sussman
Title: "Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions"
Performance Period: 7/1/26 – 3/31/27
Total Requested: \$295,794

To Whom It May Concern:

Florida State University (FSU) is pleased to submit for consideration a response to the above-referenced Request for Application. FSU's Office of Sponsored Research Administration has reviewed and approved this proposal and is prepared to negotiate and enter into any subsequent agreement resulting from this response.

FSU considers the proposed research to be fundamental research and anticipates there will be no restrictions on publication of scientific and technical information resulting from the project and no access and dissemination controls protecting information resulting from the research.

If you have any questions or require additional information, please do not hesitate to contact Alexander Karabedian via email at SRA-Pre@fsu.edu.

Thank you for your consideration.

Sincerely,

Kathy Pennington for
Stacey Patterson, VPR
Dr. Stacey Patterson
Vice President for Research

Digitally signed by Kathy
Pennington for Stacey Patterson,
VPR
Date: 2026.04.29 08:36:27 -04'00'

874 Traditions Way, Third Floor • Tallahassee, Florida 32306-4166
(850) 644-5260 • FAX (850) 644-1464



Camron Proctor

Manager, Thermal Fluids Modeling and Simulation

April 14, 2026

To Whom it May Concern:

Sandia National Laboratories (Sandia) is pleased to participate with Prof. Mark Sussman (Department of Mathematics at Florida State University) in their response to DOE's Request for Applications "The Genesis Mission: Transforming Science and Energy with AI" (DE-FOA-0003612) for topic 21, "Artificial Intelligence in Fluid Flow for Energy Components and Technologies." The title of the proposal is "Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions."

Mark Sussman has requested Sandia's unique capabilities to be included in their response to this Request for Applications. Subject to available funding and resources, Sandia anticipates participating in mutually agreed tasks, which will be performed by staff in the Thermal/Fluids Science and Engineering Department. The objective will be to develop AI-enhanced sharp-interface numerical methods that can reduce the computational cost in R&D of advanced combustion devices (such as jet engines and rotating detonation rockets). This activity will take place using existing DOE-developed simulation codes. Sandia staff will provide critical domain knowledge to anchor the choice of the user cases, while also assisting in the setup and postprocessing of the simulations. Crucially, the staff will test the AI-augmented workflow in comparison with existing best practices.

Sandia National Laboratories is a government-owned, contractor-operated facility operated by National Technology & Engineering Solutions of Sandia, LLC (NTESS) for the U.S. Department of Energy (DOE)/National Nuclear Security Administration (NNSA) under Management and Operating (M&O) Contract DE-NA0003525. As a DOE M&O Contractor, NTESS is obligated to offer its unique services to entities that wish to acquire these services using a DOE-approved agreement mechanism. Providing Sandia's services to Prof. Mark Sussman is subject to DOE review and approval in accordance with DOE policies and regulations. These services may be acquired under a DOE-approved bilateral contract with Sandia post award. Sandia implements internal training that aligns with research security training guidelines and is consistent with 42 USC 19234 and DOE Order 486.1A.

This information is provided in accordance with applicable DOE policy regarding Sandia's participation in federal government announcements. The Sandia point of contact for this potential collaboration is Marco Arienti, Sandia National Laboratories, P.O. Box 969, Livermore CA 94551.

Sincerely,
Camron Proctor, Ph.D.

Sandia National Laboratories

P.O. Box 969, MS 9042
Livermore, CA 94551

Phone: 925-294-2160
Email: cproct@sandia.gov